

# Omar Khaled

## Data Scientist | Artificial Intelligence Engineer

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### SUMMARY

Data Scientist and Artificial Intelligence Engineer focused on developing applied AI and data analytics solutions across machine learning, deep learning, computer vision, NLP, predictive analytics, and data visualization. Experienced in collecting, cleaning, interpreting, and modeling data using Python, SQL, TensorFlow, OpenCV, Spark MLlib, and statistical analysis workflows. Published AI research in IEEE and MDPI venues, with projects involving algorithms, business insights, process automation, and stakeholder-ready dashboards.

### EXPERIENCE

#### AI Engineering & Data Science Intern | [Tech Care](#) (May 2026 – Jul 2026)

- Completed a 130-hour AI internship covering data processing, data mining, model training, fine-tuning, and RAG.
- Contributed to supervised AI projects by preparing data and improving model workflows with engineers.

#### Application Development and Integration Intern | [EJADA](#) (Aug 2025 – Sep 2025)

- Developed Java backend components for process integration using REST APIs, microservices, and database workflows.
- Implemented API management, testing, and integration tasks for a final development project.
- Debugged backend workflows and improved application development structure.

#### Machine Learning & Frontend Intern | [CodeAlpha](#) (Aug 2024 – Sep 2024)

- Completed a 2-track internship covering data preprocessing, model evaluation, and frontend data visualization.
- Built ML projects using Python, data preprocessing, model training, and evaluation.
- Developed responsive web interfaces using HTML, CSS, and JavaScript.

### EDUCATION

#### Arab Academy for Science, Technology and Maritime Transport Alamein, Egypt

Bachelor of Science in Artificial Intelligence, Data Science Specialization

(2023 – 2026)

GPA: 3.93/4.00 | Excellent with Honors | Ranked 1st in Data Science Department

Graduation Project Grade: Excellent

Graduated: Aug 2026

### PROJECTS

#### [OmniRAG Assistant](#) | Python, Streamlit, Ollama, FAISS, LLaVA, RAG (Feb 2026)

- Built a multimodal RAG assistant supporting 6 input types: PDF, CSV, arXiv, image, audio upload, and live microphone.
- Implemented 2 chat modes and a 5-step retrieval pipeline using FAISS, Sentence-Transformers, and local LLMs.
- Added summarization, text-to-speech, audio transcription, and PDF chat export for non-technical users.

#### AI Vision Module for Robotics System | Python, OpenCV, TensorFlow, Computer Vision, Robotics (Sep 2025)

- Developed an AI vision module for robotic perception using object detection, achieving 94% accuracy.
- Built deep learning inference workflows for autonomous navigation, reaching 18 FPS for real-time robot decisions.
- Optimized inference workflows for lightweight edge deployment, reducing latency by 21%.

#### [ReviewPulse: Amazon Sentiment Analytics](#) | Spark MLlib, Kafka, MongoDB (Jun 2025)

- Developed a real-time sentiment analytics pipeline for large-scale Amazon reviews using Spark MLlib and Kafka.
- Processed 2,000,000 Amazon review rows and trained sentiment models on 816,926 balanced samples.
- Compared 4 ML models and achieved 81.88% accuracy and 0.8187 F1 using optimized Logistic Regression.
- Visualized real-time customer feedback insights through a Plotly Dash dashboard for stakeholder reporting.

#### [Driver Monitoring System](#) | TensorFlow, Computer Vision, Gradio (Apr 2025)

- Built a deep learning system to classify distracted driving behavior across 10 driver behavior classes.
- Compared 4 deep learning models with ensemble inference, confidence scoring, risk assessment, and prediction logging.

### TECHNICAL SKILLS

- Programming Languages:** Python, SQL, Java, JavaScript, C++
- AI & Machine Learning:** Machine Learning, Deep Learning, RAG, LLMs, Predictive Analytics, Algorithms, Model Development
- Statistics & Mathematics:** Statistical Analysis, Mathematical Modeling, Linear Algebra, Calculus, Classification, Regression
- Data Science & Visualization:** Data Cleaning, Data Preprocessing, Feature Engineering, Data Interpretation, Data Visualization, Business Intelligence
- Computer Vision & NLP:** Computer Vision, Image Classification, Object Detection, NLP, Sentiment Analysis, Text Classification
- Libraries & Tools:** TensorFlow, OpenCV, Scikit-learn, Pandas, NumPy, Spark MLlib, FAISS, Plotly Dash, Streamlit, Gradio, Jupyter
- Databases & Backend:** MySQL, SQLite, MongoDB, REST APIs, Git, GitHub

**PUBLICATIONS** 

- O. Khaled et al., “EASE-PVNet: Robust Periocular Identity Verification Across Pre- and Post-Operative Facial Images,” Machine Learning and Knowledge Extraction, 2026.  
DOI: [10.3390/make8060169](https://doi.org/10.3390/make8060169)
- O. Khaled et al., “Deep CNN and SimpleRNN Architectures for Bi-Handed Arabic Alphabet Hand-Gesture Recognition,” ICEENG, 2026.  
DOI: [10.5281/zenodo.20161073](https://doi.org/10.5281/zenodo.20161073)
- O. Khaled et al., “An Intelligent Web-Based Multi-Agent Framework for Automated University Timetabling via Constraint-Aware Optimization,” ICMISI, IEEE, 2026.  
DOI: [10.5281/zenodo.20159940](https://doi.org/10.5281/zenodo.20159940)
- O. Khaled, Z. Azzaz, E. Khatab, and O. Shalash, “Deep Learning for Post-Rhinoplasty Face Recognition,” Pattern Recognition, revised manuscript under second-round review.

**CERTIFICATIONS**

|                                                                                        |            |
|----------------------------------------------------------------------------------------|------------|
| • <a href="#">Databases and SQL for Data Science with Python</a>   IBM, Coursera       | (Jul 2024) |
| • <a href="#">AI Essentials</a>   Google, Coursera                                     | (Jul 2024) |
| • <a href="#">Prompt Engineering</a>   Dubai Future Foundation and Dubai Centre for AI | (May 2025) |
| • <a href="#">Data, Data, Everywhere</a>   Google, Coursera                            | (Jul 2024) |

**LEADERSHIP & VOLUNTEERING**

|                                                                                                         |                       |
|---------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Team Leader</b> , AAST RALLY Competition   AASTMT Society                                            | (Apr 2023 – Sep 2025) |
| • Led team planning, coordination, and project progress follow-up.                                      |                       |
| <b>Team Volunteer</b> , ECPC Egyptian Collegiate Programming Contest                                    | (Jul 2024)            |
| • Supported participant guidance, logistics, and competition execution.                                 |                       |
| <b>Technical Event Volunteer</b> , RoboCup Junior Egypt                                                 | (Apr 2025)            |
| • Supported technical activities, operations, and participant guidance.                                 |                       |
| <b>HR Member</b> , Entrepreneurship Society Alamein                                                     | (Mar 2023 – Jan 2026) |
| • Supported internal communication, member coordination, follow-up, and team engagement across 2+ years |                       |

**REFERENCES**

**Dr. Omar Shalash**

- Associate Professor, Information Technology
- Department of Artificial Intelligence, College of Engineering, Ajman University
- Artificial Intelligence Research Center, Ajman University
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**Dr. Hany Said**

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**Dr. Esraa Khatab**

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**Eng. Yasmine Khatab**

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